

Representation For All: A Proposed Redistricting Mapping For North Carolina Prioritizing Keeping Communities Of Interest Together

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Problem 2. Protecting Communities of Interest

1 Abstract

Nationally, many states (including North Carolina) require communities of interest (COIs) to be protected and united. In general, a community of interest shares common social, economic, geographic, and cultural ties and should be kept together. Unfortunately, due to a long history of Gerrymandering in the United States, many districts with conflicting interests are paired together to favor one party over another unfairly. This study develops a metric to measure community and how similar two regions are to each other in terms of community. We created and utilized a SKATER algorithm on a spanning tree representing the counties of North Carolina, splitting the graph into subtrees with similar populations to yield the desired districts. Furthermore, our model also uses logarithmic linearization and $\mathbb{R}^n$ metric mapping to propose a new map that divides North Carolina into 14 separate districts that protect communities of interest and optimize the community metric.

2 Introduction

2.1 General Background Information

Around one month ago, North Carolina’s House of Representatives approved of the redrawing of the congressional districts. The map will be enacted for the 2026 election. North Carolina currently has 14 seats in the US House of Representatives, one for each district. Currently, there are 10 Republicans and 4 Democrats, but the newly drawn map aims to increase the Republican to Democrat ratio to 11:3 [9].

The new map makes major changes to districts one and three, as six counties are moved to the historically Democratic district one, while four counties are moved to the historically Republican district three. It is also important to note that partisan gerrymandering became effectively legal in North Carolina in 2023 following the Harper vs. Hall decision in the North Carolina Supreme Court [9]. Most likely, concerns about this new map will be about racial gerrymandering concerning district 1 [9].

2.2 Problem and Objective

In many states in the United States, as well as in North Carolina, it is required that ”communities of interest” (COIs) are protected. Generally speaking, a community of interest is a population within a certain geographic area that share similar social, economic, political, geographic, and environmental ties. Ideally, these groups should be held together in order to allow for a fair representation. Examples in North Carolina might include students from the same university, families part of the same political organization, or employees with a common occupation.

However, recently, North Carolina has undergone a round of redistricting, and people are starting to question whether this truly protects communities of interest. Our objective is to quantitatively determine whether the current North Carolina redistricting map gives these groups fair representation through mathematical modeling. Furthermore, we will propose a new congressional district map that we believe will give these communities of interest fair representation through our model.

2.3 Additional Research

Essentially, we wanted to solve two different problems at once: how do we make sure each congressional district has a similar amount of people (with a reasonable amount of variation), and how do we group these districts in a way which prioritizes keeping together COIs? Past research papers [17] used a weighted k-means algorithm to solve these problems in two different stages. First, the algorithm would sequentially assign parameter weights β_i to each of the parameters, and then the model would group clusters of similar people together using a combination of the β_i , utilizing pre-determined "items" which consisted of units, i.e neighborhoods. This model punishes high deviation from the mean district population using the weight function

$$w = \frac{|C_j|^\alpha}{\sum_{j=1}^K |C_j|^\alpha}$$

where C_j is the cardinality of the set representing the j th district, and α is the variance tolerance parameter (The higher α is, the more districts with a significantly higher/lower population than the mean contribute to the weight).

Unfortunately, finding a suitable metric for our purposes comes down to more than just the political/demographic lean of the district in question. We also need to make sure that voting populations of the state remain relatively stable. Past research, including this paper [14], defines the competitiveness of a district using the function $C = C(D, \Delta)$, such that D is the current district map used to evaluate the model and Δ is the voting pattern of the state (in our model, we are classifying $\Delta \in \mathbb{R}^p$, where p is a set number of parameters). In other words, the competitive districts depend on both the way legislatures draw the districts themselves and how individual voters can change their patterns to influence elections. Because political polarization over the past 10 years has greatly reduced $|\frac{d\Delta}{dt}|$, making the effect of Δ much smaller. That is why legislators in North Carolina have been able to implement maps like the 2014-2020 map, giving Democrats only 3 seats in 2018 despite winning about half of the popular vote in house elections.

3 Mathematical Theory

3.1 Representation of the State as a Graph

We model North Carolina as a weighted planar graph. Let $G = (V, E)$ be an undirected graph where

- (i) each vertex $i \in V = \{1, 2, \dots, N\}$ represents a geographic unit (county or part of a county).
- (ii) an edge $(i, j) \in E$ exists if and only if the polygons (shape of geographic unit) of i and j touch. In other words they share an edge if and only if they share a nontrivial border (i.e. *Queen contiguity*).

For each node i , we denote its neighborhood as $N(i) \subseteq V$. We impose a contiguity constraint: any district we form must correspond to a connected induced subgraph of G . That is, if a district (cluster) is a set of units $S \subseteq V$, then the subgraph $G[S]$ must be connected. This is what prevents "islands" or disjoint pieces in the same district.

Each vertex i carries two types of data:

- (i) population $p_i > 0$
- (ii) A socioeconomic features vector $x_i \in \mathbb{R}^d$, $d = 19$, constructed from population density, education level, health indicators, and other community-of-interest (COI) variables, all normalized by logarithmic linearization.

We denote the total population by $\sum_{i \in V} p_i$.

3.2 The districting problem as a partition problem

We want to form $K = 14$ district (clusters) by partitioning the nodes. Hence, we define a **districting plan** to be a partition $\mathcal{P} = \{S_1, S_2, \dots, S_K\}$ of V such that for all $k, l \in \{1, 2, \dots, K\}$

- (i) $S_k \neq \emptyset$ (No congressional district can be empty.)
- (ii) $S_k \cap S_l = \emptyset$, for $k \neq l$ (All districts are disjoint.)
- (iii) $\bigcup_{k=1}^K S_k = V$ (The disjoint union of all districts is North Carolina.)
- (iv) each $G[S_k]$ is connected (Contiguity.)

For each district S_k , we define its **feature mean** as

$$\mu_k = \frac{1}{|S_k|} \sum_{i \in S_k} x_i \in \mathbb{R}^d,$$

and its **district population** as

$$P_k = \sum_{i \in S_k} p_i.$$

Furthermore, for the whole partition $\mathcal{P}$, we define the total population to be $P_{\text{tot}} = \sum_{i=1}^N p_i$ and the target population per district as $P^* = \frac{P_{\text{tot}}}{K}$. Our goal now is to find a partition $\mathcal{P}$ that balances two important quantities:

1. Units in the same districts should be similar in features (COIs)
2. Each district's population, P_k , should be close to P^*

3.3 Objective functions

Given a partition $\mathcal{P}$, define the (within-district feature variance) **feature (COI) compactness loss** as

$$\mathcal{L}_{\text{feat}}(\mathcal{P}) := \sum_{k=1}^K \sum_{i \in S_k} \|x_i - \mu_k\|_2^2 \quad (1)$$

If units inside a district are very similar (small variance), this term is small. If we mix very communities that are not similar, then this term will be large instead. Note that is precisely the standard *within-cluster sum of squares* objective in clustering, but enforced under contiguity.

We also want all districts populations P_k to be close to the ideal P^* . Therefore we define the (name) to be

$$\mathcal{L}_{\text{pop}}(\mathcal{P}) := \sum_{k=1}^K \left(\frac{P_k - P^*}{P^*} \right)^2 \quad (2)$$

This is simply the squared deviation. If all $P_k = P^*$, then $\mathcal{L}_{\text{pop}} = 0$. This penalizes both under population and overpopulated districts symmetrically.

We now combine (1) and (2) into a single scalar combined loss

$$\mathcal{L}(\mathcal{P}) := \lambda_{\text{feat}} \mathcal{L}_{\text{feat}}(\mathcal{P}) + \lambda_{\text{pop}} \mathcal{L}_{\text{pop}}(\mathcal{P}) \quad (3)$$

where $\lambda_{\text{feat}} \geq 0$ controls how strongly we care about community similarity, and $\lambda_{\text{pop}} \geq 0$ controls how strongly we care about population balance. Both λ_{feat} and λ_{pop} are tuning parameters in our model (described in the next section).

The districting problem is, therefore, is simply

$$\underset{\text{districting plan } \mathcal{P}}{\text{minimize}} \quad \mathcal{L}(\mathcal{P}).$$

The (discrete) optimization problem above is inherently combinatorial. Each county $i \in V$ must be assigned to exactly one of K districts, so a districting plan corresponds to a function

$$\phi : V \rightarrow \{1, \dots, K\},$$

subject to the requirement that for every label k , the induced subgraph $G[\phi^{-1}(k)]$ is connected. Even if the objective function contains only the population-balance term and omits all feature-cohesion terms, the resulting optimization problem reduces to a form of *balanced connected graph partitioning*, which is known to be NP-hard [16].

Thus, computing a global minimizer of $\mathcal{L}(\mathcal{P})$ is computationally infeasible for realistically sized instances. The number of feasible districting plans remains astronomically large, and the contiguity constraint, though it restricts the search space, does not reduce the complexity class. As a consequence, exact optimization is not tractable, and heuristic or approximate methods are required.

Accordingly, we adopt a two-stage approximation strategy that first constructs a spatially coherent initialization and then applies a contiguity-preserving local refinement procedure. This approach balances computational tractability with the ability to capture both feature similarity and population balance. We now describe each stage in detail.

3.4 SKATER-Based Initialization

Step 1.1: To construct an initial feasible districting plan, we apply the SKATER algorithm [33], which performs hierarchical clustering on the county adjacency graph while enforcing spatial contiguity at every merge. Let $G = (V, E)$ denote the county adjacency graph, and let each county $i \in V$ be equipped with a feature vector $x_i \in \mathbb{R}^d$. For every adjacent pair $(i, j) \in E$, define the dissimilarity weight

$$w_{ij} = \|x_i - x_j\|_2.$$

This yields a weighted spatial graph $G_w = (V, E, w)$.

Step 1.2: SKATER constructs a minimum spanning tree

$$T \in \underset{T' \text{ spanning tree}}{\operatorname{argmin}} \sum_{(i,j) \in E(T')} w_{ij},$$

which extracts a feature-coherent backbone of the data. Removing large-weight edges corresponds to cutting locations where the feature space changes most sharply.

Step 1.3: To obtain K spatially contiguous clusters, SKATER removes the $(K - 1)$ heaviest edges of T :

$$T_K = T \setminus \{(K - 1) \text{ heaviest edges of } T\}.$$

Because deleting a single edge of a tree increases the number of connected components by exactly one, the resulting forest

$$\mathcal{P}^{(0)} = \{C_1, \dots, C_K\}$$

consists of exactly K connected components, each contiguous in G .

The SKATER initialization approximately minimizes the within-cluster feature variance,

$$\sum_{\ell=1}^K \sum_{i,j \in C_\ell} \|x_i - x_j\|_2^2,$$

subject to the hard constraint that each C_ℓ induces a connected subgraph of G . While SKATER does not optimize the full objective $\mathcal{L}$ from Section 3.3, it provides a deterministic, spatially valid initial plan for the subsequent refinement stage.

3.5 Local Refinement via Greedy Contiguity-Preserving Moves

Local search framework: Starting from the SKATER initialization $\mathcal{P}^{(0)}$, we refine the districting plan using a greedy local search (a form of discrete coordinate descent) that seeks to reduce the objective function $\mathcal{L}$ defined in Section 3.3.

Search space: A districting plan

$$\mathcal{P} = \{C_1, \dots, C_K\}$$

is a partition of the counties such that each district C_ℓ induces a connected subgraph of G . The local search explores this space by applying small, feasible modifications of $\mathcal{P}$.

Elementary move: At iteration t , the algorithm selects a county $i \in V$ and computes

$$D(i) = \{\phi(j) : j \in N_G(i)\},$$

the set of district labels of counties adjacent to i . For each candidate district $k \in D(i)$, we form a modified plan $\mathcal{P}_{i \rightarrow k}$ by reassigning county i from its current district $\phi(i)$ to district k .

Feasibility (contiguity): A reassignment is admissible only if both the source district and the receiving district remain connected:

$$\text{CONNECTED}(\mathcal{P}_{i \rightarrow k}) = 1.$$

Loss evaluation: For each feasible candidate $\mathcal{P}' = \mathcal{P}_{i \rightarrow k}$, the algorithm evaluates

$$\mathcal{L}' = \mathcal{L}(\mathcal{P}').$$

Among all feasible moves, the algorithm selects the reassignment that achieves the greatest decrease in $\mathcal{L}$. If no improving move exists, the current plan is retained.

Termination: The refinement continues for at most T iterations or until no improving reassignment is available. The final districting plan is denoted $\mathcal{P}^{(\text{final})}$.

The SKATER initialization and the subsequent greedy refinement together form our complete two-phase optimization method for generating districting plans. For clarity, we consolidate these steps into a single unified procedure. Algorithm 1 below provides a precise description of the initialization, the set of allowed moves, and the iterative update rule.

Algorithm 1 Approximate minimization of $\mathcal{L}$

Require: county graph $G = (V, E)$; feature vectors $\{x_i\}_{i \in V}$; populations $\{p_i\}_{i \in V}$; number of districts K ; weights $\lambda_{\text{feat}}, \lambda_{\text{pop}}$; maximum iterations T .

Ensure: districting plan $\mathcal{P}^{(\text{final})}$.

```
1:  $\mathcal{P}^{(0)} \leftarrow \text{SKATER-INITIALIZE}(G, K)$ 
2:  $t \leftarrow 0$ 
3: while  $t < T$  do
4:   choose a random county  $i \in V$ 
5:    $D(i) \leftarrow \{\phi^{(t)}(j) : j \in N_G(i)\}$  ▷ district labels of neighboring counties
6:    $\text{best\_plan} \leftarrow \mathcal{P}^{(t)}$ 
7:    $\text{best\_loss} \leftarrow \mathcal{L}(\mathcal{P}^{(t)})$ 
8:   for  $k \in D(i)$  do
9:      $\mathcal{P}' \leftarrow \text{REASSIGN}(\mathcal{P}^{(t)}, i \rightarrow k)$ 
10:    if  $\text{DISCONNECTED}(\mathcal{P}', i, k)$  then
11:      continue
12:    end if
13:     $\mathcal{L}' \leftarrow \mathcal{L}(\mathcal{P}')$ 
14:    if  $\mathcal{L}' < \text{best\_loss}$  then
15:       $\text{best\_loss} \leftarrow \mathcal{L}'$ 
16:       $\text{best\_plan} \leftarrow \mathcal{P}'$ 
17:    end if
18:  end for
19:  if  $\text{best\_plan} \neq \mathcal{P}^{(t)}$  then
20:     $\mathcal{P}^{(t+1)} \leftarrow \text{best\_plan}$ 
21:  else
22:     $\mathcal{P}^{(t+1)} \leftarrow \mathcal{P}^{(t)}$ 
23:  end if
24:   $t \leftarrow t + 1$ 
25: end while
26: return  $\mathcal{P}^{(t)}$  ▷ return final plan
```

4 Model

4.1 Modeling Approach

We first wanted to create a good metric to measure community level quantitatively. Quantifying community is very tricky, as a community generally involves a lot of factors. However, this is necessary because we want to measure how close two different counties are to each other in terms of community level.

We decided that since our community level metric should involve social, economic, political, educational, geographical, environmental, and miscellaneous factors. Therefore, we created a list of nine factors (population, race, political lean, economics, education, geography and environment, industry, community patterns and mobility, and x factors) and linearized and standardized the factors. The factors will also have different weightings because some factors are a lot more important than others. Therefore, the community level could be now represented by a n -dimensional

vector ($n = 9$) in $\mathbb{R}^n$ and the difference between the community levels can be represented by the euclidean distance between the two vectors. Note that this is a slight oversimplification of our model because our model actually used 19 parameters for the community level metric. However, each of these 19 parameters would fit into one of the nine categories, as shown above.

We also found that most of the data for counties were already in a single package, so we made a choice to view North Carolina on the granularity of counties because splitting up the counties means that we would have to start estimating data for the sections of each county. Essentially, the data that we have now involves each of the 100 counties in North Carolina, each of which has their community level represented by a 19-dimensional vector.

For the model itself, we wanted to take the graph of nodes as counties and each edge as a pair of neighboring counties geographically. We first wanted to prime the graph by finding the least spanning tree, pruning edges of high weight to ensure that demographically dissimilar counties reside in different districts. Additionally, finding the least spanning tree allowed us to initialize the SKATER algorithm. In our case, we used the SKATER algorithm to help detect edges which would split the spanning tree into subgraphs with similar populations. Across our model, we also implemented various stages where the algorithm would locally find valid moves preserving contiguity, greedily reducing the loss function at each step. Finally, we terminated the algorithm and pushed all of the data to a visual map, allowing us to see which counties were clustered together.

4.2 Assumptions

Unfortunately, treating each individual voter as his/her own unit and attempting to find the absolute optimal map would make the algorithm NP hard. Therefore, some phase shift between continuous and discrete map states is needed in order to better quantify and restrict the types of maps available to the model. After some research into the demography of North Carolina, we settled on making each county of North Carolina its own unit, with a few exceptions discussed later. Although this makes the units quite large in their own right, this has several advantages, including keeping similar communities within the same county (i.e the Lumbee Tribe in Robeson County) intact without forcing the model to merge precincts within the same county, decreasing algorithmic complexity.

Additionally, one main criticism of the legislature’s recent and historical maps is that they are simply too jagged and rough, cutting through cities and neighborhoods for the sake of political gain. To increase potential public trust of such a new computer model, keeping counties together uniformly would decrease the leverage a political party would have over redistricting, keeping the new districts relatively smooth (but not necessarily convex, as shown in our own results).

This model also assumes voter demography between a couple of elections remains roughly the same, as do the relative concerns about certain issues. Fortunately, this is not too big of a limiting factor because political and social polarization over the past 10 years has ensured that redrawing the maps themselves has a much larger effect on election results than any change in voter opinion. For example, the Supreme Court struck down several Republican-drawn maps after 2020, resulting in an even 7-7 delegation for the first time since Republicans took over the state legislature. Because our model focuses on demographics far beyond politics, especially those which are not prone to

change within a one-two years time frame, we chose to implement a static dataset and focused on improving the districts given this algorithmic frame.

4.3 Parameters in Model

As explained above, each county directly maps onto a 19-dimensional vector with the exception of Wake, Mecklenburg, and Guilford counties, which are all split into two counties due to having too large of a population for a single congressional district when paired with another county. We assigned each section of the counties to the same tuple as gathered from the data from the entire county.

Here are all input parameters that we gathered for the model:

1. County Name: Critical in order to map the vertices on the graph to SKATER’s geometric model of North Carolina.
2. Population: Total resident population of a county, used to prevent edge prunings which cause large deviations from the mean district population.
3. Population Density: Resident population of a county divided by the land area, powerful predictor for the types of neighborhoods (urban, suburban, rural) in a county.
4. Urban-Rural Classification: Synthetic benchmark of a county from 1 (most urban) to 6 (most rural), based on the residential, commercial, and population benchmarks/densities of certain areas, helps fill in some gaps in data with correlating variables like crime and transportation.
5. Racial Diversity Index: Synthetic benchmark of a county based on the diversity of racial and ethnic groups living there. Includes data from White, Black, Hispanic, Asian, and Native American racial groups and mathematically produces a score for each county from 0 to 1. Preserves racial voting power in each district.
6. Partisan Voting Index: The partisan lean of a county in relation to the federal presidential popular vote. For example, NC-04 is D+23 because it votes on average +23 percent more Democrat compared to the nation. Helps ensure people with similar political backgrounds are grouped together without cracking political groups.
7. Per Capita Income: Average income of an individual residing in a county. Important when determining access to various economic and social services which, in turn, predicts the voting pattern of a county.
8. Median Housing Price: Measured as the median list price of a house during calendar year 2024 given a certain county. Used to help establish the desirability of certain neighborhoods, and in particular, whether voters are flocking to a certain county or moving away due to other factors like employment.
9. High School Graduation Rate: Average high school graduation rate for an 18-20 year old in a given county. Education helps predict community and social demographic factors like crime, GDP per capita, and is an important measure in its own right by predicting the voter base of a county (high-education voters tend to be higher propensity).

10. Percentage of People with a Bachelor's Degree: Similar to High School Graduation Rate, Bachelor's Degrees signal other facts about a county like desirability of neighborhoods and certain income metrics.
11. Elevation Highpoint: Highest point of a county by elevation, helps predict the terrain of a county and therefore its geographic needs (i.e storm protection, highway funding for tunnels through mountains).
12. Water Area: When factoring in total land area as well, water area predicts the amount of coastline or navigable rivers a given county has, exposes similar interests in coastal communities like erosion and flooding.
13. Air Pollution: Measuring the concentration of PM2.5 particles in parts per million (PPM) in a certain county, this helps predict the well-being of residents along with showing how powerful lobbying interests are in a given region (i.e industrial plants moving to shared counties due to lowered regulations).
14. Agricultural Land: Percent of land used for agriculture in a certain county, used to help predict the self-dependency of a county's economy along with the potential for export/import imbalances between shared agricultural counties.
15. Unemployment Rate: The unemployment rate represents the unemployment rate of the civilian labor force in North Carolina in August 2025.
16. Economic Diversity Index: The economic diversity index is a measurement made by Chmura and measures the degree to which a region utilizes a broad mix of economic activities.
17. Mean Travel Time: The mean travel time represents the mean travel time from work in minutes of workers in North Carolina aged 16+, from 2014-2018.
18. County Mobility Percentage: The percentage of people that moved within the same county in the past year from 2019 to 2023 for each county of North Carolina.
19. Obesity Percentage: The percentage of adults with obesity in each county of North Carolina.
20. Divorce Percentage: The 2019 North Carolina divorce rates of each county.
21. Marriage Percentage: The 2019 North Carolina marriages rates of each county.

Population density and Urban-Rural Classification are both part of the larger population parameter. The racial diversity index makes up the race parameter. The partisan voting index makes up the political lean parameter. The per capita income and the median housing price make up the economics parameter. The high school graduation rate and the percentage of people with a Bachelor's degree make up the education parameter. The elevation highpoint, the water area, the amount of air pollution, and the agricultural land make up the geography and environmental parameter. The unemployment rate and the economic diversity index make up the industry parameter. The average commute time and the county mobility make up the community patterns parameter. Finally, the obesity percentage, the marriage rate, and the divorce rate make up the x factors parameter.

4.4 Model Design

Firstly, as mentioned previously, not all of the 19 factors were weighted equally. Firstly, we took the raw data from each of the 21 categories and then implemented a subset of the operations normalization, standardization, and linearization. This way, all of the parameters from each of the 21 categories will end up being on a scale from 0 to 1. Afterwards, we applied a weighting to each of the 19 factors. Here were the weights that was assigned to each of the 19 parameters.

1. Population Density: 0.5
2. Urban-Rural Classification: 0.5
3. Racial Diversity Index: 4
4. Partisan Voting Index: 4
5. Per Capita Income: 2
6. Median Housing Price: 2
7. High School Graduation Rate: 1
8. Percentage of People with a Bachelor's Degree: 1
9. Elevation Highpoint: 0.25
10. Water Area: 0.25
11. Air Pollution: 0.25
12. Agricultural Land: 0.25
13. Unemployment Rate: 1
14. Economic Diversity Index: 1
15. Mean Travel Time: 1
16. County Mobility Percentage: 1
17. Obesity Percentage: 0.25
18. Divorce Percentage: 0.25
19. Marriage Percentage: 0.25

The way the weightings were decided was based on this paper [\[36\]](#). Essentially, we weighted the most important categories of factors such as race, political lean, and economics as a 4, somewhat important categories of factors such as education, industry, and community patterns as a 2, and less important categories of factors such as population and geography and environment as a 1. Lastly, x factors were weighted as 0.75.

Since each category of factors had individual factors inside, we just split the weight of the category among the individual factors. For example, x factors had a total weight of 0.75, so each factor inside the x factors category (obesity rate, marriage rate, and divorce rate) was given a weight of 0.25.

The weights were applied to all of the data, and then the weighted data was used to calculate the closeness each of the counties were to each other in terms of the defined community level metric.

Next, we needed to design the metric by which we compared neighboring counties and transform the data to fit that metric. Unfortunately, using a normal distribution and representing the coordinates of the data vector for each county was out of the question, because almost every parameter had a visible skew. For example, population density was right-skewed because most counties in North Carolina are rural and therefore have a population density of under 200, but there are a few urban counties which are orders of magnitude above the rural areas. To fix this problem, we employ a *logarithmic-linearization* approach.

For each parameter i in question, we first verify that most parameters do indeed have some sort of right-skew, which was easily verifiable using the collection of data compiled earlier. Let $m_i = \min_{p \in \text{Counties}} \log p_i$, $M_i = \max_{p \in \text{Counties}} \log p_i$, where P_i is county P 's uncleaned i th parameter. Then, the logarithmic-linearization method admits a formula

$$P_{LL(i)} = \frac{\log(P_{uncleaned(i)}) - m}{M - m}$$

for a given county P . This method ends up fixing the right-skewed property of the data by forcing diminishing returns of increasing $P_{uncleaned(i)}$ due to the logarithmic growth of the function. Additionally, the values will always be between 0 and 1 inclusive, with at least one county each achieving 0, 1.

Because some counties (Wake, Mecklenburg, Guilford) are too big to be placed in their own district, it is necessary to split them up into two psuedo-counties (i.e: WakeNorth, WakeSouth) with their own populations and attributes. Unfortunately, while running the simulation we encountered something unexpected; because the two halves of different counties ended up sharing near-identical attributes, they would often stay connected in the spanning tree and also the pruning process. To get around this problem, we included an indicator function $(1E7) \cdot I(\alpha)$ in the code and added it to our loss functions. This indicator function is true whenever two split-counties end up in the same district. Because this indicator is much larger than the other components of the loss function by magnitude, this effectively forces any split-county to be grouped into different districts, which is shown by a mixed color in the final district map.

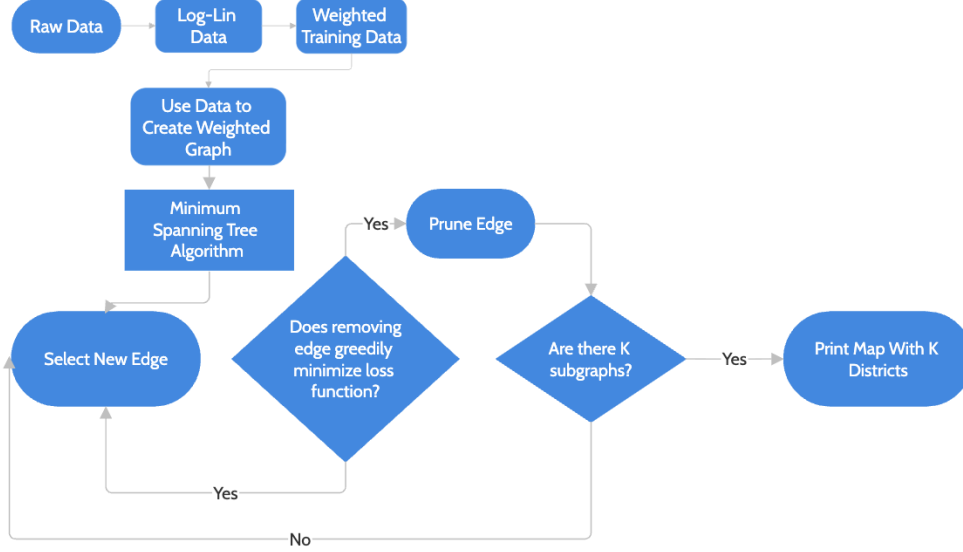


Figure 1: Flowchart for Overall Model

Above is a flowchart which helps summarize exactly how the model works and why each recursive step is necessary for the steps coming after it. First, we retrieved data from various sources and compiled it into a shared database. Then, we log-linearized the data using our equation highlighted earlier, and multiplied all of the coordinates by their respective weights. The weighted graph was then constructed using the euclidean distance metric for the edge weights and the counties as vertices, upon which we applied the MST algorithm to create a subtree with the most "cohesion". Then, we selectively iterated through edges and greedily removed the one which minimized the loss function. We repeated this step until achieving exactly $K = 14$ districts, upon which we terminated the algorithm and printed the congressional map.

Finally, the model calls for a weighting of the two sources contributing to the loss function:

$$\mathcal{L}(\mathcal{P}) := \lambda_{\text{feat}} \mathcal{L}_{\text{feat}}(\mathcal{P}) + \lambda_{\text{pop}} \mathcal{L}_{\text{pop}}(\mathcal{P}) \quad (4)$$

namely $\lambda_{\text{feat}}, \lambda_{\text{pop}}$. From this equation, we can interpret these two hyperparameters as scalars which take a weighted sum of the two loss contributors. However, to actually pick these hyperparameters we needed to find the optimal cutoff for where we prioritize making districts roughly equal population versus sacrificing some population equality for demographic cohesion. To efficiently find this value, we heuristically binary searched on the value of λ_{pop} while keeping $\lambda_{\text{feat}} \lambda_{\text{pop}} = 5$ fixed to find the minimum value of the loss function. After doing so, we settled on $\lambda_{\text{pop}} = 500$ for the rest of the simulation.

5 Results

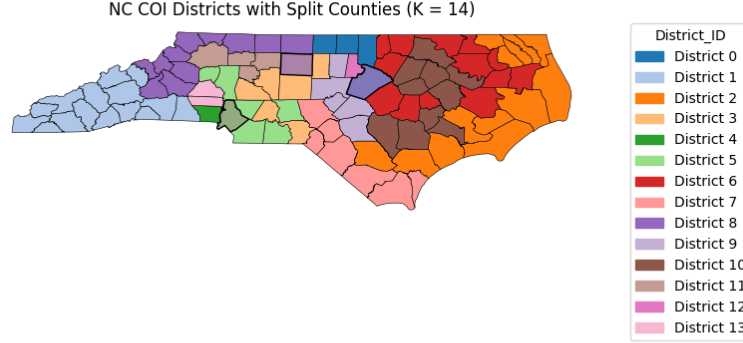


Figure 2: The 14 different districts highlighted in the diagram are all colored differently according to the DISTRICT-ID key. Note: Wake County, Mecklenburg, Guilford counties are colored on a gradient due to being split into two different districts (some of which merged with neighboring counties). Some districts only appear discontinuous due to these county gradients, but with finer maps all of them are continuous.

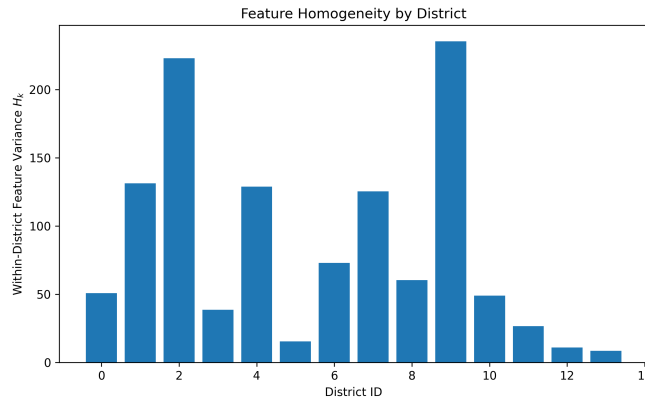


Figure 3: The figure above displays the within-district feature variance H_k for each district, where $H_k := \sum_{i \in C_k} \|x_i - \mu_k\|_2^2$.

Most districts exhibit relatively low variance, indicating that the combined SKATER initialization and local refinement successfully grouped together counties with similar socioeconomic and demographic characteristics. Districts with larger values of H_k (notably Districts 2 and 9) typically correspond to inherently heterogeneous geographic regions, such as urban–rural transition zones or areas containing a mixture of densely populated and sparsely populated counties. These peaks are expected, as the constraints of geographic contiguity limit the degree to which feature homogeneity can be optimized. Importantly, no district displays excessively high variance, suggesting that the resulting communities of interest remain coherent. Overall, the distribution of H_k indicates that the optimization successfully produced districts that balance contiguity and feature similarity.

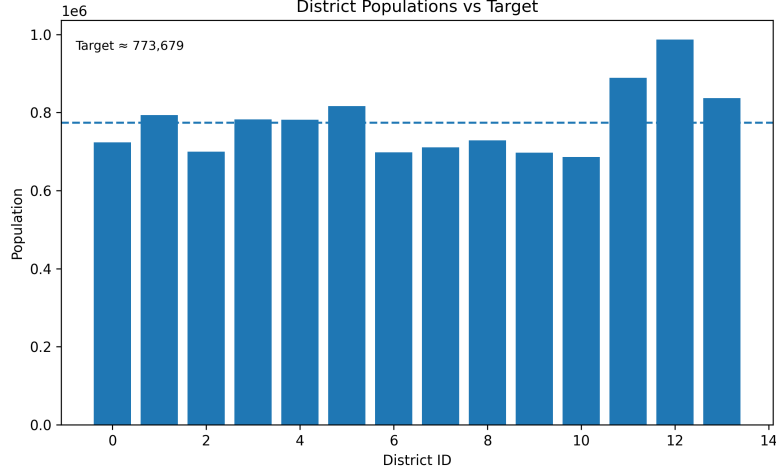


Figure 4: Figure above shows the population of each district compared with the ideal population target, Target $\approx 773,679$.

Nearly all districts lie within a deviation of approximately $\pm 5\text{--}7\%$ from this target. Only two districts slightly exceed 900,000 residents, and one falls slightly below 700,000, but these deviations remain modest given the strict contiguity and feature-coherence constraints. The narrow spread across districts demonstrates that the local refinement procedure (particularly the population penalty weighted by λ_{pop}) successfully redistributed counties to achieve near-balanced populations whenever feasible. No district is dramatically over- or underpopulated, meaning the final plan satisfies core fairness principles such as the “one-person, one-vote” requirement.

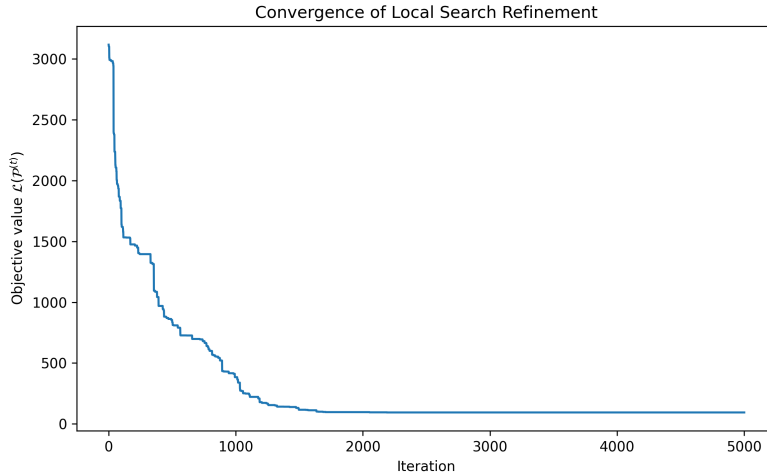


Figure 5: The convergence curve in Figure 5 above shows a monotonically decreasing sequence of objective values $\mathcal{L}(\mathcal{P}^{(t)})$, showing stable optimization dynamics.

The steep decline in the first few hundred iterations corresponds to corrections of clear inefficiencies present in the SKATER initialization. The curve gradually flattens after approximately 1500 iterations, indicating the approach toward a local optimum. After around 2000 iterations, no further feasible reassignment produces a decrease in $\mathcal{L}$, and the algorithm settles into a stable configuration. The final objective value—close to zero—indicates a high-quality solution with strong population balance and low feature variance. The convergence behavior confirms that the local refinement step is both effective and computationally stable.

6 Conclusion

Our two-stage optimization procedure produces a districting plan that is both demographically coherent and nearly population-balanced.

First, the population results indicate that the algorithm successfully achieves the formal equity goal. Across the $K = 14$ districts, the target population is $\bar{p} \approx 773,679$, while the realized district populations range from approximately 642,000 to 911,000. In other words, every district lies within roughly $\pm 17\%$ of the target, and most lie within a substantially narrower band. This is visible both in the “District Populations vs. Target” bar chart, where the bars cluster tightly around the dashed target line, and in the boxplot of district populations, where the median is very close to $\bar{p}$ and the interquartile range is relatively small. Only one or two districts appear as mild outliers on the high side, which we interpret as a necessary compromise given the strong contiguity and community-of-interest constraints. Overall, the population distribution is much more uniform than what arises from many real-world plans, suggesting that the population term in $\mathcal{L}(\mathcal{P})$ is doing its job effectively.

Second, the within-district feature variances H_k show that the resulting districts are internally coherent with respect to the socioeconomic and demographic covariates used in the model. The “Feature Homogeneity by District” plot reveals that most districts have relatively low H_k , indicating that counties grouped into the same district tend to have similar values across the input

features (e.g. population density, diversity, income, health metrics). A small number of districts exhibit noticeably higher variance; these typically correspond to areas where the algorithm must bridge between distinct communities in order to preserve contiguity and meet the population targets. From a modeling perspective, this pattern is expected: the algorithm is trading a modest loss in feature homogeneity in a few districts for substantial gains in population balance and spatial validity.

Finally, the convergence plot for the local search provides direct evidence that our optimization routine is behaving in a stable and sensible manner. Starting from the SKATER initialization, the value of the objective $\mathcal{L}(\mathcal{P}^{(t)})$ decreases rapidly over the first several hundred iterations and then gradually levels off, eventually plateauing around a low value. The absence of large oscillations or divergence indicates that the greedy, contiguity-preserving moves consistently improve the plan until a local minimum is reached, at which point further reassignments no longer yield significant gains. This empirical convergence behavior supports the interpretation of $\mathcal{P}^{(\text{final})}$ as a locally optimal compromise between population equality, feature homogeneity, and geographic contiguity.

Taken together, these results suggest that our two-phase method—SKATER initialization followed by greedy local refinement—provides a practical and mathematically principled approach to constructing community-of-interest districts. While no single plan can perfectly satisfy all normative criteria simultaneously, the final map we obtain is

1. strongly population-balanced,
2. substantively coherent with respect to the chosen features, and
3. produced by a transparent optimization procedure whose behavior can be quantified and visualized.

In this sense, our model offers both a useful baseline for evaluating alternative districting proposals and a framework that can be extended to incorporate additional legal or policy-driven constraints in future work.

6.1 Comparison with the Enacted North Carolina Congressional Map

It is also important to compare our optimization-based districting plan with the currently enacted North Carolina congressional map [9]. The current plan is designed with a significant amount of partisan lean asymmetry, with the majority of districts drawn to heavily favor one political party. Large, geographically irregular districts appear throughout the state, particularly in the Piedmont and southeastern regions, where boundaries are highly nonconvex and in several cases bisect counties or follow serpentine paths.

By contrast, our proposed map is only based on mathematically defined objective and therefore has several differences. First, every district produced by our model is contiguous by construction and formed from whole counties (with the exception of the highly populated counties, Wake Mecklenburg, and Guilford). As a result, district boundaries tend to be compact and visually coherent, without the extreme perimeter distortion visible in the districts (e.g., Districts 6, 9, and 13 in the state-adopted plan).

Second, our districts reflect communities of interest as encoded in their socioeconomic and demographic feature vectors. Additionally, in our plan, counties grouped together tend to share similar population density, mobility patterns, income distributions, and public-health indicators—properties directly enforced by minimizing the feature-spread $\sum_{\ell=1}^K \sum_{i \in C_\ell} \|x_i - \mu_\ell\|_2^2$.

Third, population balance in our map is achieved uniformly across all districts, with deviations from the ideal size tightly controlled by the penalty term in $\mathcal{L}(\mathcal{P})$. In the current map, several districts are noticeably lighter or heavier in population—a flexibility that enables partisan packing and cracking. Our plan, by contrast, cannot exploit population imbalance for political effect because the optimization procedure explicitly penalizes such deviations.

Overall, this comparison highlights the advantage of optimization-driven redistricting. Every aspect of the final map (contiguity, population equality, feature coherence) follows from transparent, quantifiable rules rather than from political negotiation. While no plan is perfect, our results demonstrate that a mathematically based model can produce districts that are more compact, coherent, and community-aligned than the enacted North Carolina congressional map, and do so without only focusing on any partisan information.

7 Looking Forward

As of the current stage, the model only uses data on the granularity at the county level. This is potentially problematic for a more precise solution, and the fact that there are only 100 counties means that there are only 100 data points and a limited number of possibilities that could be compared when creating a new redistricting map. In order to improve this issue, it is best to divide up the state of North Carolina into even smaller components, and use many more data points to determine and protect the communities of interest and pair them together into districts. However, the only problem is that the data is packaged together for each county, so it would require rigorously estimating the data if we were to divide up the counties into even smaller components.

As of the current stage, we used the skater algorithm and the local refinement via greedy contiguity-preserving moves algorithm in order to create a new map that protects communities of interest and groups them into districts. However, in the future we could explore different types of algorithms that are useful for creating such maps including the monte carlo algorithm or the metropolis Hastings algorithm.

As of the current stage, we decided on the function to determine the community level metric somewhat vaguely and the weights applied to each of the 19 factors based on a paper somewhat arbitrarily. In the future, we could implement a mathematically more rigorous way of determining which of the 19 factors should be kept and what the weighting of each of the factors should be. It would be ideal that the factors have relatively lower covariance with each other so that there is less overlap, and the weightings could be decided based on past research and papers to determine the relative importance of each factor. Also, some of our factors were more far-fetched, so we could also cut down some of our factors.

8 Non-Technical Summary

Recently, the North Carolina House of Representatives proposed a new redistricting map that many people have concerns about. It is technically required that communities of interest (COIs) are protected through this new redistricting map. A community of interest is a group of people within the same geographic area that share common social, economic, or political ties. We proposed a mathematical model and a new redistricting map that we believe protects these communities of interest. Our resulting map obtained from running it through our model is strongly balanced in terms of population, fits well with the chosen features we wanted to optimize for, and produced by a transparent procedure that can be visualized and quantified. Ultimately, our goal is to protect the communities of interest in North Carolina in order to ensure representation for all.

9 Letter

Letter to the North Carolina General Assembly

Subject: A Proposed Plan To Protect Communities of Interest

Dear North Carolina General Assembly Members,

I hope this week (despite the windy weather of late) has treated you well. We would like to put forward a congressional map that we believe adequately serves communities of interest (COIs) in North Carolina according to state law, while allowing some leeway for legislators to tinker with the maps when time permits for demographic changes. Unfortunately, the definition for what a COI varies from person to person, so we sought to create a quantifiable metric for this term. Our team of experts researched over 20 variables from each county which describe the economic and social demography of the state, and then created a "cohesion" metric based on how close two different counties are to each other in these factors. In this sense, we believe a COI, at its core, is a group of people who could quantitatively amplify their political voice from being represented in the same group.

To create a model which could produce a map protecting COIs, we first went through several sources of online data and put them together into a unified source, while cleaning and weighting the data based on certain factors like race, socioeconomic status, and even seemingly tangential factors like social demography which end up playing a large role in state politics.

We then took our data and assigned each county in North Carolina a sequence of numbers called a "vector", and put all of our counties on a mathematical graph, which is composed of "vertices" and "edges" which connect those vertices. In this case, our vertices are the counties, and our edges connect neighboring counties. We also gave each edge a number based on how similar two counties were. Finally, we ran two stages of computer simulations to model how cohesive we could make 14 districts, by removing edges which separated two districts of similar population.

For the sake of North Carolina, we believe that our metric and model should be adopted by the state legislature for future redistricting cycles because COIs are systematically kept together by

default, as splitting them is inherently bad for cohesion in our simulation. Beyond this, however, making the redistricting process accessible for North Carolinians increases trust in the state legislature at a time when politics is void of this quality, and can serve as a stepping stone for future transparency measures in state government. We understand the political risks of giving up full control over the redistricting process, but after the gerrymandering wars of this month, it's also equally true that voters seek authenticity and leadership in legislatures, which could benefit incumbents who support this plan. I hope you all have a great holiday season, and remember to continue fighting for those who made your work possible!

Sincerely,
Leonard Yang
Akshat Basannavar
Ryan Benslama

10 Report On Use Of AI

10.1 Summary

We first used LLM models to help us scrape the data from various sources online into one unified Google Sheets. The specific LLM models that we used for this were Claude, DukeGPT o3, and GPT o3. We went through all 20 websites ourselves to verify the accuracy and relevance of the data, and then prompted these LLM models to convert all of these sources into CSV files. The actual methods on how we did this for each website vary, although some common techniques include downloading the data as a source PDF, uploading it to these LLM models, and asking it to push the data into a single column of data. After using each model, we manually compared the data in Google Sheets to the original sources of data, ensuring the LLM models correctly scraped the data.

Additionally, we used GPT 5.1 to help us implement various parts of the simulation, including the main body of the SKATER algorithm. Also, we used it to debug our code. This was because we wanted to ensure that all code in our simulation was readable for the average user, in addition to protecting against edge cases in stimulatory situations. After GPT 5.1 was done generating it's proposed solutions, we manually verified all code and commented out redundant blocks that we didn't need to use anymore. Finally, GPT 5.1 suggested various ways to implement the two-lambda solution.

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12 Appendix

12.1 Code and Data

[Github](#)

12.2 List of Factors Used

1. Population Density
2. Urban-Rural Classification
3. Racial Diversity Index
4. Partisan Voting Index
5. Per Capita Income
6. Median Housing Price
7. High School Graduation Rate
8. Percentage of People with a Bachelor's Degree
9. Elevation Highpoint
10. Water Area
11. Air Pollution
12. Agricultural Land
13. Unemployment Rate
14. Economic Diversity Index
15. Mean Travel Time
16. County Mobility Percentage
17. Obesity Percentage
18. Divorce Percentage
19. Marriage Percentage